

Project Synopsis

Title: Hospital Chatbot for Color-Coded Clinical Pathways

1. Introduction (Background of Study)

"Triage is not just about sorting; it is about the right patient getting the right care at the right time." — South African Triage Group

In high-functioning healthcare systems, triage is a continuous, dynamic status that dictates a patient's entire journey through the hospital. Standardized scales, such as the South African Triage Scale (SATS) and ETAT, utilize color codes to classify clinical severity:

- **Red:** Emergency
- **Orange:** Very Urgent
- **Yellow:** Urgent
- **Green:** Non-Urgent
- **Blue:** Dead

These colors are designed to be more than just labels; they are supposed to trigger specific "Clinical Pathways" that define exactly where a patient goes and how quickly they are treated.

However, in many Low- and Middle-Income Countries (LMICs), the utility of this color often ends at the front desk. Once a patient leaves the triage area, the hospital ecosystem effectively "forgets" their acuity level. A "Yellow" patient transferred to the X-ray department or a general ward often joins a generic queue, losing the time-critical protections their color was meant to provide.

This project proposes a hospital-wide digital framework that attaches a specific, automated clinical pathway to every patient. By digitizing the triage logic, the system acts as a "Flow Management Engine," ensuring that the urgency assigned at the door dictates the allocation of resources (beds, labs, doctors) throughout the patient's entire stay.

2. Problem Statement

Current hospital workflows in resource-constrained settings suffer from a critical failure known as "Acuity Discontinuity."

- **The "Lost" Urgency:** A patient triaged as "Orange" (Very Urgent) in the Emergency Department (ED) often becomes invisible to the system once moved to a diagnostic unit or ward. There is no digital mechanism to track if their strict 10-minute review target is met.
- **Resource Mismatch:** Without automated filtering, "Green" (Non-urgent) patients frequently congest acute care zones, while "Red" (Emergency) patients face dangerous delays due to bed blockages.
- **The "Blue" Bottleneck:** In many facilities, patients who are Dead on Arrival (Blue) are managed ad-hoc. The lack of a standardized "Blue Pathway" leads to logistical chaos, occupying clinical beds and causing distress to staff and other patients.
- **Subjective Routing:** The decision of where to send a borderline patient is often left to individual staff judgment, leading to variability in care and overcrowding in the ED.

3. Objectives

General Objective

To design a logic-based digital framework that automates the assignment and management of color-coded clinical pathways to optimize patient flow and safety in LMIC hospitals.

Specific Objectives

1. **To Digitize Triage Logic:** Develop a "Digital Acuity Engine" that converts standard SATS/TEWS algorithms into code, automatically calculating patient color based on vital signs and symptoms.
2. **To Map Hospital-Wide Pathways:** Define strict, step-by-step logistical and clinical rules for each of the five color categories (Red, Orange, Yellow, Green, Blue) beyond the Emergency Room.
3. **To Design the "Virtual Interface":** Create a chatbot-based user interface that communicates these pathways to staff (e.g., "Patient deteriorating, re-assess now") and patients (e.g., "Proceed to Zone B").
4. **To Simulate System Efficiency:** Model the theoretical impact of this

automated routing on Patient Throughput and Resource Utilization compared to traditional manual workflows.

4. Methodology

The research will follow a Design Science Research (DSR) approach, structured in four phases:

Phase 1: Knowledge Capture & Workflow Analysis

- Reviewing Standard Operating Procedures (SOPs) from partner hospitals (e.g., Komfo Anokye Teaching Hospital) to understand current manual flows.
- Interviewing triage nurses to identify "pain points" where the manual color-coding system fails.

Phase 2: Logic Modeling (The "Artifact")

Constructing the decision trees (Algorithms) for the Five-Path Logic:

- **Red Path:** Immediate Resuscitation.
- **Orange Path:** High Dependency Admission (10-min timer).
- **Yellow Path:** Parallel Processing (Auto-ordering labs while waiting).
- **Green Path:** Decongestion (Diversion to Outpatient Polyclinic).
- **Blue Path:** Dignity Protocol (Direct transfer to Mortuary services).

Phase 3: Prototype Development

- Developing a functional prototype of the Chatbot Interface using a conversational AI framework (e.g., Dialogflow or Python). This will serve as the input/output terminal for the Acuity Engine.

Phase 4: Simulation Testing

- Running historical patient data (anonymized) through the digital model to verify if it assigns the correct color and route compared to expert physician judgment.

5. Analysis and Expected Results

Analysis

The system's performance will be analyzed using Confusion Matrices to measure:

- **Sensitivity:** Did it catch the sick patients?
- **Specificity:** Did it correctly filter the healthy ones?
- **Process Efficiency:** Measured by simulating the reduction in "Time to Provider."

Expected Results

- A standardized "Digital Architect" for hospital flow that functions 24/7 without fatigue.
- A significant theoretical reduction in Emergency Department congestion by successfully diverting "Green" patients.
- Improved safety and dignity through the dedicated management of "Red" and "Blue" pathways.

6. Conclusion and Recommendations

Conclusion

Transforming triage from a manual "gatekeeping" task into a digital "navigation" system addresses the root causes of ED overcrowding and patient neglect. By enforcing color-coded pathways, hospitals can ensure that safety is systemic, not accidental.

Recommendations

- Hospitals in LMICs should adopt digital acuity tools to support (not replace) overburdened staff.
- Future policy should integrate "Blue Pathways" explicitly into emergency training to handle mortality with dignity and efficiency.

7. References

1. Rominski, S., et al. (2014). The implementation of the South African Triage Score (SATS) in an urban teaching hospital, Ghana. *African Journal of Emergency Medicine*.
2. Kamau, S., et al. (2024). Comparison between the Smart Triage model and ETAT guidelines in triaging children in Kenya. *PLOS Digital Health*.
3. Nurchis, M., et al. (2025). Assessing the Impact of Waiting Time on Triage Color Code Assignment and One-Year Mortality. *Health Science Reports*.

4. Mitchell, R., et al. (2023). Systematic review: What is the impact of triage implementation on clinical outcomes in LMIC emergency departments?
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